Two-Branch Attention Adversarial Domain Adaptation Network for Hyperspectral Image Classification

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***Abstract*— Recent studies have shown that deep domain adaptation (DA) techniques have good performance on cross-domain hyperspectral image (HSI) classification problems. However, most existing deep HSI DA approaches directly use deep networks to extract features from the data, which ignores the detailed information of HSI in spectral and spatial dimensions. To effectively exploit the spectral–spatial joint information for DA of HSIs, we propose a two-branch attention adversarial DA (TAADA) network in this article. In the TAADA network, a two-branch feature extraction (TBFE) subnetwork is first designed as a generator to extract the attention-based spectral–spatial features. Then, a discriminator based on two classifiers with the multilayer FC-BN-ReLU-Dropout structure is constructed. Based on adversarial learning between the generator and the discriminator, the ability of discriminative feature extraction and cross-domain classification is improved simultaneously. Finally, the TAADA network can adjust the distribution between the source and target domains and extract domain-invariant features. Experimental results on three cross-scene HSI classification tasks show that our proposed TAADA outperforms some existing DA methods.**

***Index Terms*— Adversarial network, domain adaptation (DA), hyperspectral image (HSI), two-branch.**

I. Introduction

In RECENT years, the classification of hyperspectral images (HSIs) has become one of the hot research issues [1], [2], and it has been widely used in environmental and disaster monitoring, agriculture, and military. The classification of HSI is difficult due to the large amount of data, high dimensionality, and inaccessibility of labels [2], [3]. In practical situations, we usually face a more complicated cross-domain HSI classification problem where the existing HSI for training the classification model and an unknown HSI that needs to be classified come from different scenes or domains and have different distributions. The distribution inconsistency problem may be caused by the differences in acquisition environment conditions or/and sensors, which greatly aggravates the difficulty of HSI classification. To produce robust cross-domain HSI classification, many domain adaptation (DA)-based classification methods are proposed [2], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13].

The traditional DA methods are roughly carried out from the following aspects [2]: 1) domain-invariant feature selection; 2) data distribution adaptation; and 3) classifier adaptation. Among them, the feature transformation and distribution adaptation methods are widely researched, such as subspace alignment (SA) [14], [15], geodesic flow kernel (GFK) [16], transfer component analysis (TCA) [17], transfer joint matching (TJM) [6], [18], joint distribution adaptation (JDA) [19], correlation alignment (CORAL) [20], joint geometrical and statistical alignment (JGSA) [8], [21], locality-preserving joint transfer (LPJT) [22], and discriminative cooperative alignment (DCA) [10]. These traditional DA methods aim to learn linear or nonlinear transformations such that the subspaces or distributions of the transformed source and target domains are aligned. However, they usually focus on hand-crafted features, and their ability to learn features from data adaptively is limited. Recently, deep-learning-based methods have been widely used in automatical feature learning for DA.

Unlike hand-crafted features in the traditional DA methods, the deep learning methods can automatically learn features using deep neural networks. Most of the current deep DA methods add an adaptation layer to the original deep network architecture to realize the adaptation of different domains. The deep DA methods can be roughly divided into discrepancy-based methods [23], [24], [25], [26], adversarial-based methods [27], [28], [29], and others [30]. Long et al. [23] extended deep convolutional neural networks (CNNs) to DA scenarios and proposed a deep adaptation network (DAN). DAN adopts the multikernel maximum mean difference (MMD) metric to measure the domain's distribution difference. Based on DAN, Zhu et al. [24] proposed a

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deep subdomain adaptation network (DSAN) using the idea of subdomain adaptation, which learns a transport network by aligning the correlated subdomain distributions between different domains based on the local MMD. To incorporate the idea of CORAL into a deep learning architecture, Sun and Saenko [25] proposed a DeepCORAL method. Zhu et al. [26] proposed a multirepresentation adaptation network (MRAN), which performs cross-domain classification tasks by aligning multiple feature representations. The network can effectively use different types of information and expand the MMD to conditional MMD to compute the class-conditional distribution differences of two domains. There are also many DA networks based on adversarial learning. Adversarial training, inspired by generative adversarial network (GAN) [31], is a useful approach to learn domain-invariant features. Long et al. [27] proposed a joint adaptation network (JAN), which adopts an adversarial training strategy to adjust the joint distribution between domains based on the joint MMD criterion. Yu et al. [28] proposed a dynamic adversarial adaptation network (DAAN), which learns domain-invariant representations dynamically and assesses the global and local domain distributions simultaneously. Tzeng et al. [32] proposed an adversarial discriminative DA (ADDA) method, which includes three steps, i.e., pretraining, adversarial adaptation, and testing. The model first pretrained a source encoder CNN using source samples and labels, and then learned a target encoder CNN in the adversarial learning step such that the discriminator can distinguish between the source and target features and the target encoder can fool the discriminator to classify the target domain into the source domain, thus further enabling the classifier trained on the source features to accurately classify the target features. Saito et al. [29] proposed a deep DA method based on the maximum classifier discrepancy (MCD) that leverages task-specific decision boundaries and adversarial learning ideas to adjust the distribution of domains.

Recently, many deep DA methods have been introduced for cross-domain HSI classification [7], [9], [33]. Wang et al. [9] applied deep adversarial DA to cross-domain HSI classification and proposed a domain adversarial broad adaptation network (DABAN). DABAN performs cross-domain classification by adding a DA layer to the traditional deep network and uses an adversarial learning strategy to reduce the distribution differences (i.e., marginal and conditional distribution differences, and second-order statistic difference) between domains and to learn domain-invariant features. Zhang et al. [7] proposed a topological structure and semantic information transfer network (TSTnet) based on the topological relationship between hyperspectral pixels. TSTnet is divided into two branches after extracting features using CNN, one of which will explore the topological relationship of planar pixels of HSIs to reduce the domain shift using subgraph structure, and the other will measure the domain shift of the extracted features using MMD, and finally the consistency loss of both is included in an optimization objective to achieve a robust effect in the cross-domain HSI classification task. However, these methods directly use CNNs to extract hyperspectral features, ignoring the detailed features in the spatial and spectral dimensions of HSI. Considering the differences in features in different spectral bands and different spatial locations, some scholars applied the attention mechanism in spectral and spatial feature learning and achieved excellent results for HSI classification [34], [35], [36], [37].

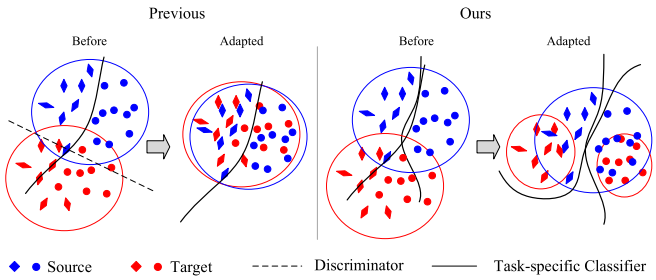


Fig. 1. General idea of dual classifier discriminator.

Considering the superior performance of adversarial learning in DA and the attention-based spatial–spectral feature representation ability of HSI, in this article, we propose a two-branch attention adversarial DA (TAADA) network to simultaneously extract spatial–spectral features and eliminate distribution discrepancy between different HSIs. Similar to the original GAN model, the proposed TAADA also consists of a generator and a discriminator. However, in the generator, a two-branch feature extraction (TBFE) subnetwork is constructed to extract attention-based spatial–spectral features of HSIs. The discriminator consists of two classifiers with a multilayer FC-BN-ReLU-Dropout structure to learn discriminative features.

The discriminator in our TAADA is different from the general GAN-based DA networks. In the general GAN-based DA networks [31], [32], [38], [39], the discriminator aims only to distinguish whether the samples are from the source or target domain without considering the category of samples. It needs another task-specific classifier built on the source domain to perform the final classification task. The discriminator and task-specific classifier are two separated parts, where the discriminator simply tries to align distribution and does not consider the relationship between target samples and decision boundary. Hence, the generator is likely to generate ambiguous features near the classification decision boundary [29]. As shown in the left side of Fig. 1, after adversarial learning, the discriminator can make the source and target features aligned. However, the task-specific classifier produces classification confusion in the decision boundary (i.e., three target samples near the decision boundary are misclassified). To align distribution and simultaneously perform domain distinguishing and class discriminating, we design a new discriminator to replace the previous discriminator and task-specific classifier. The new discriminator consists of two classifiers, which can function as a discriminator to distinguish the source and target features by maximizing the output of two classifiers and meanwhile can function as a classifier to classify target samples. As shown in the right side of Fig. 1, the new discriminator can detect target samples that are far from source samples and align their distributions by minimizing the classification disagreement between two classifiers.

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The contributions of the proposed TAADA can be summarized as follows.

1) Combining adversarial learning and DA, we propose a TAADA network, which can realize the classification of HSIs in the target domain using only the labeled samples in the source domain. We embed DA into the process of adversarial learning, and the final features are likely to be both discriminative and invariant to the change in domains. The proposed network is simple but very effective and achieves the state-of-the-art performance.
2) Based on the characteristics of HSIs, we design a TBFE subnetwork to extract the squeeze-excite (SE) attention-based spatial–spectral joint features of HSIs. The subnetwork focuses on the detailed features, which is beneficial for the subsequent training of the classifier. Unlike many previous DA-method-based fixed feature representations, we combine DA and deep feature learning within an adversarial training process. The features are dynamically adjusted based on the generator. We provide a detailed analysis on the effect of features. Although the SE-based attention mechanism is very simple, the final results are relatively good. Other attention mechanisms can also be considered.
3) We design a discriminator consisting of two classifiers F_1 and F_2 of a specific multilayer FC-BN-ReLU-Dropout structure and use the output differences in the two classifiers for target samples to train the generator. The discriminator not only can detect the target samples but also can perform specific classification tasks. Different from previous Dropout-FC-BN-ReLU classifier in MCD [29], our classifier has a multilayer FC-BN-ReLU-Dropout structure and is more effective in alleviating the "variance shift" problem especially in the DA tasks. In addition, the multilayer structure can further improve the classification performance.

II. Proposed Method

This article focuses on the unsupervised DA problem, where a source domain D_s has labels and a target domain D_t has no labels. Denote $X_s = \{\mathbf{x}_1^s, \ldots, \mathbf{x}_{n_s}^s\}$ and $Y_s = \{y_1, y_2, \ldots, y_{n_s}\}$ as the sample and label set in the source domain, respectively, where $\mathbf{x}_i^s \in \mathcal{R}^d$ and $y_i^s \in \{1, 2, \ldots, C\}$. Denote the unlabeled target samples in D_t as $X_t = \{\mathbf{x}_1^t, \ldots, \mathbf{x}_{n_t}^t\}$.

Fig. 2 shows the flowchart of the proposed TAADA network, which contains two parts: a generator and a discriminator. As a generator, TBFE contains two spectral and spatial branches. The discriminator consists of two classifiers F_1 and F_2, which are specific classifiers with i layers and each layer is organized in the order of FC-BN-ReLU-Dropout.

The adversarial process between the generator G and the discriminator D (i.e., F_1 and F_2) is divided into multiple alternative steps, as shown in the bottom green dashed box of Fig. 2. When the generator TBFE (G) is fixed, the attention-based spatial–spectral features can be extracted by the TBFE subnetwork. Two classifiers F_1 and F_2 should classify the source samples correctly and produce target classification results $p_1(\mathbf{y}|\mathbf{x}_t)$ and $p_2(\mathbf{y}|\mathbf{x}_t)$. As the initial two classifiers are different, there exists classification discrepancy [i.e., the difference between the outputs of two classifiers, $d(p_1(\mathbf{y}|\mathbf{x}_t), p_2(\mathbf{y}|\mathbf{x}_t))$] on target samples, as shown in the shaded area in the first subfigure. The goal of adjusting classifiers F_1 and F_2 is to detect the target samples that are far from the source samples. Hence, F_1 and F_2 (i.e., discriminator) are updated by maximizing the classification discrepancy $d(p_1(\mathbf{y}|\mathbf{x}_t), p_2(\mathbf{y}|\mathbf{x}_t))$ on target samples, as shown in the second subfigure where the shaded area is enlarged. In the next step, F_1 and F_2 are fixed, and the generator G is optimized to fool the discriminator. That is, G is updated by minimizing the classifiers' discrepancy on the target samples. With the decrease in classification discrepancy, the target samples will move toward the source samples, and the source and target features are gradually aligned, as shown in the third subfigure. By alternatively updating the generator and classifiers, and gradually adjusting the source and target distributions, finally TAADA can accurately classify the target samples.

A. Generator: Two-Branch Feature Extraction

For the cross-domain HSI classification problem, the traditional CNN-based feature extraction methods may lose part of the information due to the existence of rich spectral and spatial information in HSI data. To better use the information of HSIs, we design a TBFE subnetwork that uses two branches to extract the spatial and spectral features of HSIs separately. For simplicity, both the branches use the most conventional operations: 2-D and 3-D convolution, normalization and pooling, and add the classical SE attention module at the end of them to obtain features with stronger representation ability.

The operation process is illustrated in Fig. 2. Taking the Houston data as an example, we explain how TBFE performs data feature extraction. During training, we set the window size as 7×7, that is, patch_size $= 7$.

1) Spectral Feature Extraction: This part introduces the spectral feature extraction branch (i.e., orange arrow in Fig. 2). This branch consists of two convolutional layers, a residual block, and a SpectralSE structure that squeezes the spatial information and excites the spectral features [40]. The sample matrix is extracted according to patch_size and has the size of $7 \times 7 \times 48$. First, the data are processed using 24 convolution kernels of $1 \times 1 \times 7$ size, and this step focuses on the extraction of spectral dimensional features. After three convolutions and one residual operation, the obtained data are put into the SpectralSE module to get attention-based spectral features. The detailed structure of the SpectralSE is shown in Fig. 3. Similar to the traditional SE model [41], SpectralSE considers the importance of each spectral band, and then squeezes and excites different bands. Let $\mathbf{V} = [\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_C]$ denote 2-D spatial patches, where $\boldsymbol{v}_k \in \mathbb{R}^{H \times W}$ and $H = W = 7, C = 128$. The SpectralSE operation is

$$z_k = \mathbf{F}_{sq}(\boldsymbol{v}_k) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \boldsymbol{v}_k(i, j), \quad k = 1, \ldots, C \tag{1}$$

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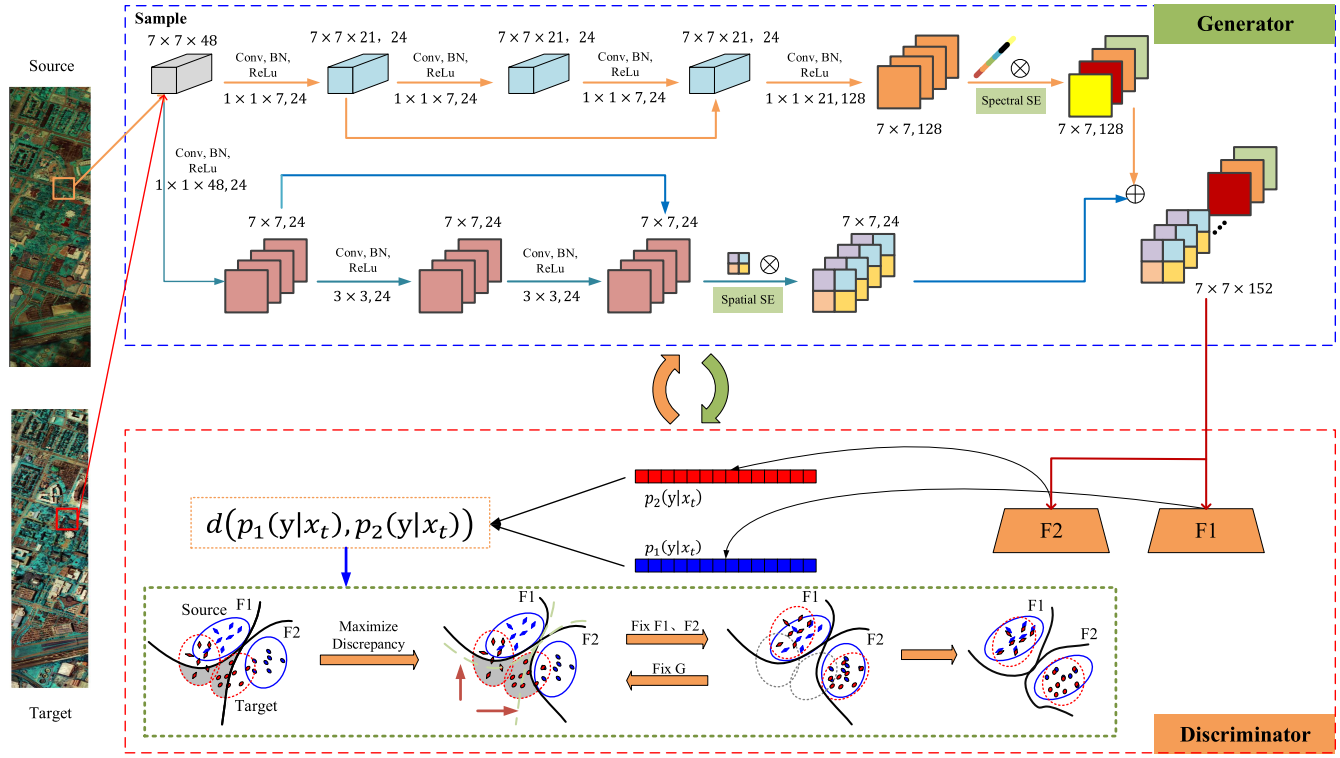


Fig. 2. Flowchart of the TAADA network; the generator part is in the blue dashed box and the discriminator part is in the red dashed box.

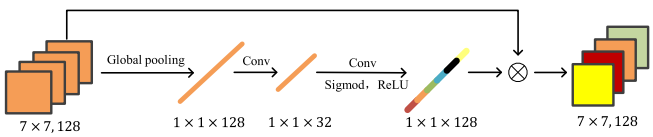


Fig. 3. Structure of SpectralSE.

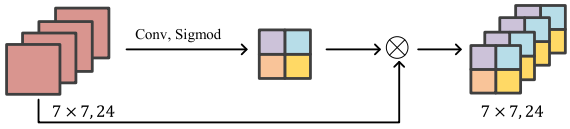


Fig. 4. Structure of SpatialSE.

where $F_{sq}(\cdot)$ is called as the squeeze operator. This operation is followed by an activation operation. The sigmoid activation function $\sigma(\cdot)$ is used to obtain the final output as

$$s = \mathbf{F}_{ex}(\mathbf{z}, \mathbf{W}) = \sigma(W_1(\delta(W_2 z))) \tag{2}$$

where $\delta(\cdot)$ is the ReLU function, and $\mathbf{W_1} \in \mathbb{R}^{C\times(C/2)}$ and $\mathbf{W_2} \in \mathbb{R}^{(C/2)\times C}$ are the weight matrixes of the two fully connected (FC) layers.

With the SpectralSE, the spectral features are extracted while preserving the original spatial information, so that each patch of the output features can better represent the spectral information.

2) Spatial Feature Extraction: The spatial feature extraction branch consists of two convolutional layers, a residual block, and a SpatialSE structure that squeezes the spectral information and excites the spatial features, as shown in blue arrow in Fig. 2. First, the data are processed using 24 $1 \times 1 \times 48$ convolutional kernels, focusing on spatial information extraction. Then, the data are put into the SpatialSE structure after two convolutions and one residual operation. The structure of the SpectralSE is shown in Fig. 4. All channel information is squeezed first, and then excited and scaled according to the original spatial information. Let $\mathbf{V} = [\boldsymbol{v}^{1,1}, \boldsymbol{v}^{1,2}, \ldots, \boldsymbol{v}^{i,j}, \ldots, \boldsymbol{v}^{W,H}]$ denote the 49 $1 \times 1 \times 24$ vectors (convertible to 24 7×7 2-D patches), where $\boldsymbol{v}^{i,j} \in \mathbb{R}^{1\times1\times C}$ and $H = W = 7, C = 24$. The SpatialSE operation is expressed as

$$\boldsymbol{q} = \mathbf{F}_{ex}\big(\mathbf{F}_{sq}(\mathbf{V})\big) = \sigma(\boldsymbol{W} \otimes \mathbf{V}) \tag{3}$$

where $\mathbf{W} \in \mathbb{R}^{1\times1\times C}$ and $\mathbf{q} \in \mathbb{R}^{W\times H}$.

3) Combination of Spatial and Spectral Features: In the previous SE-based HSI classification method [40], the SE module is directly performed on the original HSI data and the obtained spectral and spatial attention features have the same size, and the final feature in the linear combination of spectral and spatial features. Here, we perform the SE-based attention module on the spectral and spatial convolutional features. As the number of feature channels is different (i.e., $7 \times 7 \times 128$ and $7 \times 7 \times 24$), a concatenation operator is performed to obtain the final features. That is, the output of the spectral branch is connected in series with the output of the spatial branch, and the combined feature has the size of $7 \times 7 \times 152$. Finally, after average pooling of size 7×7, the final feature is a 152-D vector. In the above feature extraction

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processes, the SE-based attention module can also be replaced by other attention modules. In the experiments, we will show that even the classical and simple SE module can generate much better results.

B. Discriminator: Two Classifiers

The discriminator D consists of two classifiers F_1 and F_2. Unlike the traditional discriminators in GAN, the discriminator D not only can detect target samples but also can perform a specific target domain classification task.

Different from the Dropout-FC-BN-ReLU classifier in MCD [29], the classifier in our network has a multilayer FC-BN-ReLU-Dropout structure, as shown in Fig. 5, where FC, BN, and ReLU represent the fully connected, batch normalization, and activation function layer, respectively. In the design of the classifier, BN provides the desired distribution for the activation layer, so BN is usually before the ReLU and after the FC layer [42]. As for the order of dropout, previous research showed that dropout changes the variance of a specific neural unit when transferring the state of the network from training to testing (i.e., "variance shift") [43]. The "variance shift" leads to unstable numerical behavior in inference, which ultimately leads to poor network performance when dropout is applied before BN [43]. In the DA problem, the source and target domain data inherently suffer from inconsistent data distribution. That is, from source training to target testing, there exists great variance shift. So, dropout cannot be applied before the BN [44], [45], and we adjust the classifier structure as the order FC-BN-ReLU-Dropout. To further improve the classification performance, a multilayer FC-BN-ReLU-Dropout structure is designed. For input feature $\mathbf{x}$ generated by the TBFE subnetwork, the classifiers F_1 and F_2 generate C-dimensional probabilistic vectors as $p_1(\mathbf{y}|\mathbf{x})$, $p_2(\mathbf{y}|\mathbf{x})$, and then obtain C-dimensional logical vectors by applying a softmax function, as shown in Fig. 5.

Although the classifier is simple, it is very effective for the cross-domain classification problems, as shown in the experimental parts. The proposed TAADA uses a combination of two classifiers to become a discriminator, but in practice, multiple classifiers can also be used to form a discriminator.

C. Two-Branch Attention Adversarial Domain Adaptation Network

The proposed TAADA aims to use the output difference in two classifiers to adjust the sample distribution between the source and target domains while extracting the spectral–spatial joint features of HSIs. The training steps are divided into three steps: 1) network pretraining; 2) training discriminator D (classifiers F_1 and F_2); and 3) training generator G.

1) Network Pretraining: The network pretraining is to enable the generator G to extract discriminative features of the source domain and the classifiers F_1 and F_2 to accurately classify the source domain samples. The purpose of this step is achievable because sample labels of the source domain are available. The objective of the network pretraining is to minimize the cross entropy loss as

$$\min_{G,F_1,F_2} \mathcal{L}(X_s, Y_s) = -\mathbb{E}_{(\mathbf{x}_s,y_s)\sim(X_s,Y_s)} \sum_{c=1}^{C} \mathbf{1}_{[c=y_s]} \log p_1(\mathbf{y}|\mathbf{x}_s) - \mathbb{E}_{(\mathbf{x}_s,y_s)\sim(X_s,Y_s)} \sum_{c=1}^{C} \mathbf{1}_{[c=y_s]} \log p_2(\mathbf{y}|\mathbf{x}_s) \tag{4}$$

where $\mathbf{1}_{[c=y_s]}$ is the indicator function.

2) Training Discriminator D: In this step, we fix the generator G trained in the previous step and train the discriminator D. By training the discriminator, we increase the output difference in two classifiers to detect the target domain samples that are not supported by the source domain information. We measure the difference using the absolute value metric as

$$d\big(p_1(\mathbf{y}|\mathbf{x}_t), p_2(\mathbf{y}|\mathbf{x}_t)\big) = \big\|p_1(\mathbf{y}|\mathbf{x}_t) - p_2(\mathbf{y}|\mathbf{x}_t)\big\|_1 \tag{5}$$

where $p_1(\mathbf{y}|\mathbf{x}_t)$ and $p_2(\mathbf{y}|\mathbf{x}_t)$ denote the probability output of two classifiers on target sample $\mathbf{x}_t$, respectively.

We update D (classifiers F_1 and F_2) with the following objective function:

$$\min_{F_1,F_2} \mathcal{L}(X_s, Y_s) - \mathcal{L}_{\text{adv}}(X_t) \tag{6}$$

where $\mathcal{L}_{\text{adv}}(X_t)$ measures the classification discrepancy between two classifiers on the target samples as

$$\mathcal{L}_{\text{adv}}(X_t) = \mathbb{E}_{\mathbf{x}_t\sim X_t}\big[d(p_1(\mathbf{y}|\mathbf{x}_t),\ p_2(\mathbf{y}|\mathbf{x}_t))\big]. \tag{7}$$

3) Training Generator G: The discriminator D (i.e., classifier F_1 and F_2) is fixed, and the generator G is optimized to fool the discriminator. That is, the generator G is optimized to minimize the output difference in two classifiers on the target samples. The objective function of this step is

$$\min_{G} \mathcal{L}_{\text{adv}}(X_t). \tag{8}$$

Finally, the optimized generator G should generate discriminative features such that the predictions of two classifiers for the target samples are as consistent as possible.

During training, these three steps are iteratively repeated to continuously update the generator G and classifiers F_1 and F_2 of the network. Its purpose is to enable the generator G to extract domain-invariant spectral–spatial features of HSI while the classifiers can accurately classify the source and target samples. The pseudocode of TAADA is summarized in Algorithm 1.

III. Experimental Results

A. Datasets

In the experiments, six HSI datasets, i.e., Houston2013 and Houston2018, Dioni and Loukia, and Hangzhou and Shanghai, are used.

Houston2013 and Houston2018: These two datasets were acquired by the Compact Airborne Spectrographic Imager (CASI)-1500 sensor over the University of Houston campus and the neighboring urban area in different years [46], [47]. The Houston2013 dataset has the size of 349 × 1905 pixels

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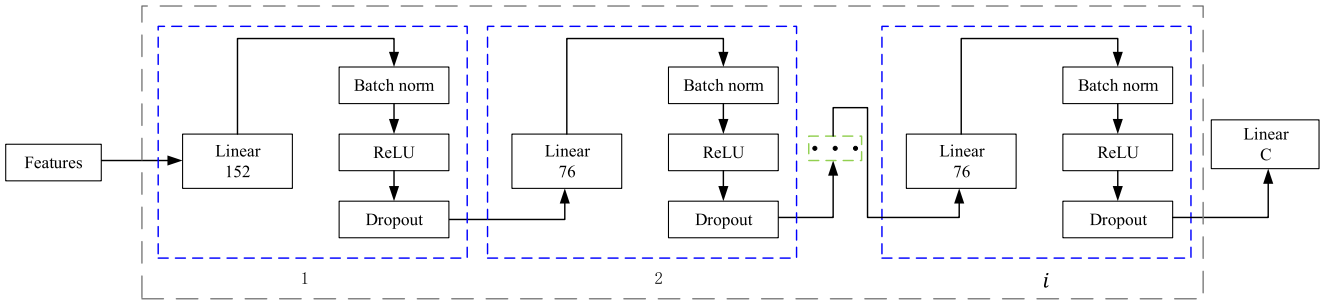


Fig. 5. Structure of classifiers F_1 and F_2.

Algorithm 1 TAADA

Input: Source sample X_s and label Y_S, target sample X_t.
Parameters: Source domain sample expansion factor $large_num$ and hyperparameter m.
Output: Predicted label Y_t for the target domain.

Train in batches according to $batch_size = 128$:
1. Extract source and target domain samples according to $patch_size$.
2. Expand the source domain samples by the source domain sample expansion factor.
3. **Repeat**:
   1) Network pretraining:
   $$(X_s, Y_s) \rightarrow \mathcal{L}(X_s, Y_s)$$
   2) Training classifiers F_1 and F_2: Fix generator G, update classifiers F_1 and F_2.
   $$(X_s, X_s, Y_s) \rightarrow (\mathcal{L}(X_s, Y_s), \mathcal{L}_{\text{adv}})$$
   3) **Repeat**: Training generator G: Fix the classifiers F_1 and F_2, and update the generator G (Repeat m times).
   $$(X_s, X_s, Y_s) \rightarrow \mathcal{L}_{\text{adv}}$$

Until convergence
4. Obtain the final label Y_t for the target domain.

and contains 144 bands in the spectral range from 380 to 1050 nm, with a total of 15 categories. The Houston2018 dataset has 48 spectral bands and contains 20 categories. For consistency, 48 spectral bands of Houston2013 are selected [7].

HyRank: The HyRank data were acquired by the Hyperion sensors. It consists of five HSIs [48], and each has 176 spectral bands. Two images, i.e., Loukia with 250 × 1376 pixels and Diono with 249 × 945 pixels, are chosen in the experiment.

Hangzhou–Shanghai: The Hangzhou and Shanghai datasets were captured by the EO-1 Hyperion hyperspectral sensor. There are 198 spectral bands in the images [49]. The Hangzhou data consist of 590 × 230 pixels, including roads, buildings, plants, West Lake, and Qiantang River basin. The Shanghai data consist of 1600 × 230 pixels, which includes roads, buildings, plants, and the water of the Yangtze and Huangpu Rivers.

Based on the six HSI datasets, the following three DA tasks are designed.

TABLE I
NUMBER OF SAMPLES IN THE HOUSTON TASK

Houston2013 and Houston2018 data			
No.	Class	Source (Houston2013)	Target (Houston2018)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

1) *Task 1 (Houston Task):* The source and target domains are chosen from the Houston2013 and Houston2018 images, respectively. Seven common classes in these two images are considered for the DA task, as shown in Table I.
2) *Task 2 (HyRank Task):* The Dioni and Loukia datasets are set as the source and target domains, respectively. Twelve common classes in these two images are considered for the DA task in Table II.
3) *Task 3 (Hangzhou–Shanghai Task):* The Hangzhou and Shanghai images are set as the source and target domains, respectively. There are three common classes in these two dataset. These common classes are considered for the DA task. The number of samples in the different domains is shown in Table III.

B. Comparison Methods

The proposed TAADA is compared with the following DA methods: KNN (no adaptation), SA [14], JDA [19], DANs [23], DSAN [24], MRAN [26], ADDA [32], MCD [29], and TSTnet [7].

After DA, the data distributions of the source and target domains can be aligned to some extent, and then the target features can be classified using a classifier trained on the features of the source domain. The final performance of each DA method is evaluated by the classification results on the target samples. For SA and JDA, the dimension of subspace is set to 20, and the classifier is 1-NN classifier. For the deep DA algorithms DAN, DSAN, MRAN, ADDA, MCD, and TSTnet, patch_size is selected from {5, 7, 9, 11, 12},

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TABLE II
NUMBER OF SAMPLES IN THE HYRANK TASK

Dioni and Loukia data			
No.	Class	Source (Dioni)	Target (Loukia)
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Vegetation	361	422
7	Dense Sderophyllous Vegetation	5035	2996
8	Sparce Sderophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421
Total		20024	10317

TABLE III
NUMBER OF SAMPLES IN THE HANGZHOU–SHANGHAI TASK

Hangzhou and Shanghai data			
No.	Class	Source (Hangzhou)	Target (Shanghai)
1	Water	18043	123123
2	Land/building	77450	161689
3	Plant	40207	83188
Total		135700	368000

TABLE IV
CLASSIFICATION ACCURACIES FOR THE HOUSTON TASK

Class	KNN	SA	JDA	DAN	DSAN	MRAN	ADDA	MCD	TSTnet	TAADA
1	78.94	78.79	81.82	63.78	26.46	58.31	54.84	65.85	**82.39**	78.46
2	**94.82**	**94.82**	80.36	75.57	78.07	74.00	74.63	87.77	77.78	84.86
3	66.52	66.56	65.80	62.80	**73.17**	67.53	63.56	55.60	64.01	66.57
4	**100**	**100**	**100**	**100**	**100**	81.82	**100**	**100**	95.45	89.55
5	64.00	64.02	60.50	46.04	41.05	55.71	58.01	85.17	82.72	**86.43**
6	68.10	68.23	59.86	69.89	72.16	69.07	73.87	72.57	76.93	**84.72**
7	60.91	60.71	72.07	73.44	61.30	58.55	62.91	**75.04**	53.71	66.60
OA(%)	69.49	69.55	64.15	67.92	67.18	66.57	70.03	74.49	75.19	**81.63**
κ	0.539	0.539	0.487	0.521	0.508	0.509	0.524	0.614	0.619	**0.706**

batch_size is uniformly set to 128, and the learning rate lr is chosen from $\{0.0001, 0.001, 0.01, 0.1\}$. In the TAADA algorithm, batch_size $= 128$, num_epoch $= 50$, and lr is selected from $\{0.01, 0.1\}$. The window size is set to 7×7, 5×5, and 7×7 in the three sets of experiments. The overall accuracy (OA) and Kappa coefficient (κ) on the target domain are used to evaluate the classification performance of each method.

C. Experiments

1) Experiments on Houston Task: In this task, the source and target domains are HSIs collected from the University of Houston and its surrounding fields in different years. Since the datasets were collected at different times, there are large differences between the two datasets, and this DA task is rather difficult.

Table IV provides the experimental results of different DA algorithms on the Houston data. It can be seen from Table IV that the classification accuracy of the deep DA algorithms DAN, DSAN, and MRAN are even lower than that of KNN (no adaptation). This shows that if the characteristics of HSIs are not fully considered, using general deep learning methods cannot guarantee the DA performance. The traditional DA algorithms SA and JDA, especially JDA, also do not classify the data well. The MCD method provides relatively better results on this task, which indicates that the adversarial idea has certain superiority for the classification of cross-domain HSIs. TSTnet shows slightly better result than MCD using graph optimal transmission (GOT) and MMD to perform graph topological relationship alignment and distribution alignment, respectively. Among all the methods, our TAADA provides the best results by extracting domain-invariant joint spatial–spectral features using the TBFE and adversarial learning, and its classification accuracy reaches 81.63%. Fig. 6 shows the classification maps of each comparison algorithm in the Houston task. It is clear that TAADA shows better results in the circled white rectangle region in the upper part of the map.

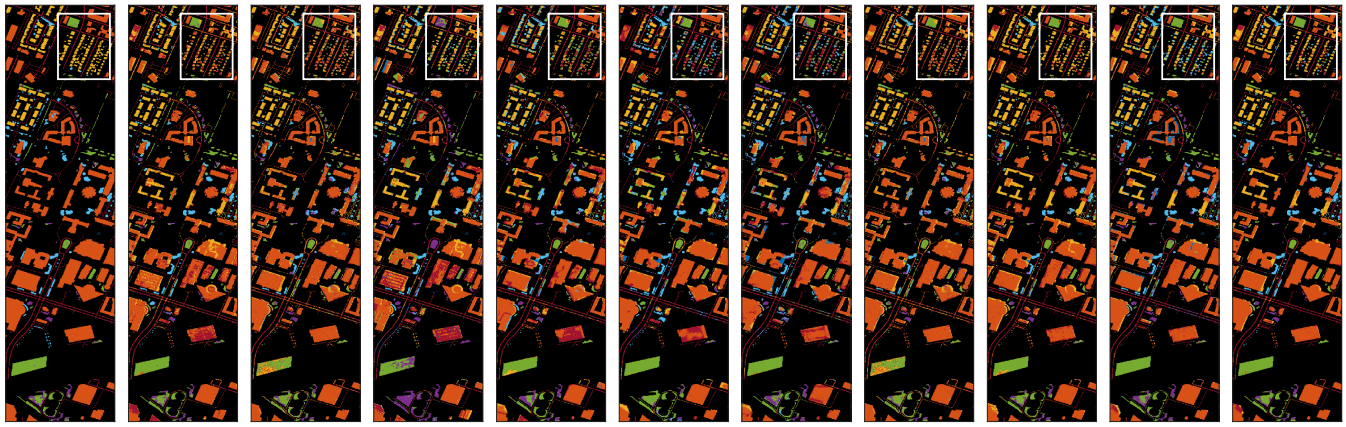
TABLE V
CLASSIFICATION ACCURACIES FOR THE HYRANK TASK

Class	KNN	SA	JDA	DAN	DSAN	MRAN	ADDA	MCD	TSTnet	TAADA
1	15.28	**22.92**	20.14	0	4.85	3.88	26.21	4.85	17.06	21.84
2	52.24	47.76	92.54	**85.19**	0	37.04	7.41	7.41	10.05	69.81
3	36.90	29.52	**75.65**	21.13	60.56	34.51	39.67	59.62	13.85	56.20
4	0	65.82	69.62	12.66	29.11	44.30	8.86	**73.42**	1.08	0
5	0.14	21.48	29.26	3.52	7.95	3.88	3.70	12.74	48.37	**57.00**
6	25.40	1.20	5.80	29.62	9.00	10.66	**34.60**	26.78	7.21	28.82
7	**80.49**	64.91	67.23	71.76	68.06	68.93	75.77	64.49	69.12	76.25
8	34.86	62.22	63.22	70.98	**76.66**	73.74	61.37	72.64	69.26	69.14
9	31.19	40.84	18.32	**54.14**	22.31	31.83	25.81	22.81	74.83	40.38
10	38.60	35.93	**40.25**	0	36.42	0.22	51.66	0.22	8.83	11.77
11	**100**	96.98	**100**	99.86	**100**	**100**	99.93	99.85	**100**	**100**
12	**100**	30.38	**100**	**100**	**100**	**100**	99.29	99.76	99.69	**100**
OA(%)	54.03	54.48	61.11	59.75	61.39	58.60	60.94	59.15	63.23	**68.03**
κ	0.436	0.447	0.529	0.501	0.524	0.488	0.516	0.507	0.549	**0.610**

2) Experiments on HyRank Task: This DA task needs to classify 12 classes in the Loukia data based on the information of the Donio data. Table V provides the experimental results of different DA algorithms. Since the number of categories is large and there are many similar categories in this dataset, such as the fourth category ("Fruit Trees") and the fifth category ("Live Groves"), the classification performance of each algorithm on this dataset is poor. Even so, TAADA outperforms other comparison algorithms. In particular, the proposed TAADA shows relatively better results on classes with large samples, such as Classes 5, 7, 8, and 11. Fig. 7 shows the classification maps of different methods in the HyRank task. The proposed TAADA shows relatively better results than other methods especially in the circled white rectangle region.

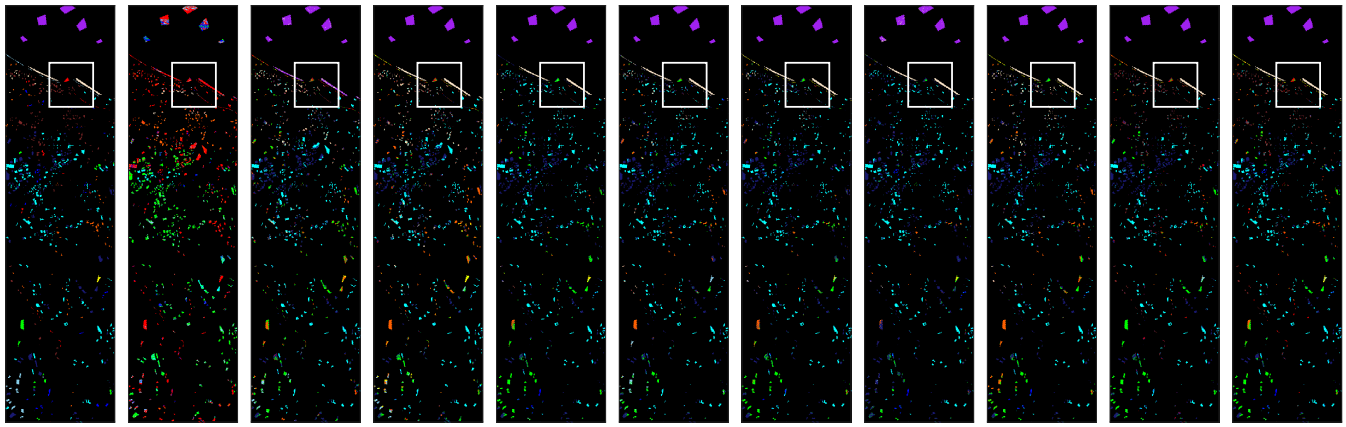
3) Experiments on Hangzhou–Shanghai Task: Table VI shows the classification results of different DA methods on the Hangzhou–Shanghai task. When directly using KNN for classification without DA, the result is poor with κ of 0.5990 and OA of 72.07%. The subspace learning method SA only makes a slight improvement in this case. JDA aligns both the marginal and conditional distributions of the source

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 6. Houston task. (a) Ground truth, (b) KNN (69.49%), (c) SA (69.55%), (d) JDA (64.15%), (e) DAN (67.92%), (f) DSAN (67.18%), (g) MRAN (66.57%), (h) ADDA (70.03%), (i) MCD (74.49%), (j) TSTnet (75.19%), and (k) TAADA (81.63%). (The white rectangle circles indicate the areas that TAADA classifies more accurately compared with the comparison algorithm.)



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 7. HyRank task. (a) Ground truth, (b) KNN (54.03%), (c) SA (54.48%), (d) JDA (61.11%), (e) DAN (59.75%), (f) DSAN (61.39%), (g) MRAN (58.60%), (h) ADDA (60.94%), (i) MCD (59.15%), (j) TSTnet (63.23%), and (k) TAADA (68.03%). (The white rectangle circles indicate the areas that TAADA classifies more accurately compared with the comparison algorithm.)

TABLE VI
CLASSIFICATION ACCURACIES FOR THE HANGZHOU–SHANGHAI TASK

Class	KNN	SA	JDA	DAN	DSAN	MRAN	ADDA	MCD	TSTnet	TAADA
1	93.35	92.71	93.38	86.88	93.48	89.44	85.88	91.48	93.63	**96.00**
2	41.51	62.47	71.58	82.27	82.45	80.34	73.84	75.58	83.74	**90.47**
3	**100.00**	85.99	99.99	99.20	99.49	99.13	87.71	99.56	99.61	98.65
OA(%)	72.07	77.91	85.29	87.64	89.99	87.63	81.00	86.32	90.64	**94.17**
κ	0.599	0.670	0.780	0.811	0.848	0.812	0.712	0.794	0.857	**0.910**

and target domains, and its classification OA is improved, reaching 85.29%. However, it still cannot classify the second category ("land/building") well. Using the deep features, the classification accuracy of the second category has been greatly improved, and the overall classification accuracy has been further improved. TAADA uses adversarial learning to fully consider the spatial and spectral information, so that the classification accuracy on this DA task reaches a high level of 94.17%. Fig. 8 shows the classification maps of each comparison algorithm in the Hangzhou–Shanghai task. From the map, we can clearly see that the proposed TAADA method provides the best results in the center circled white rectangle region.

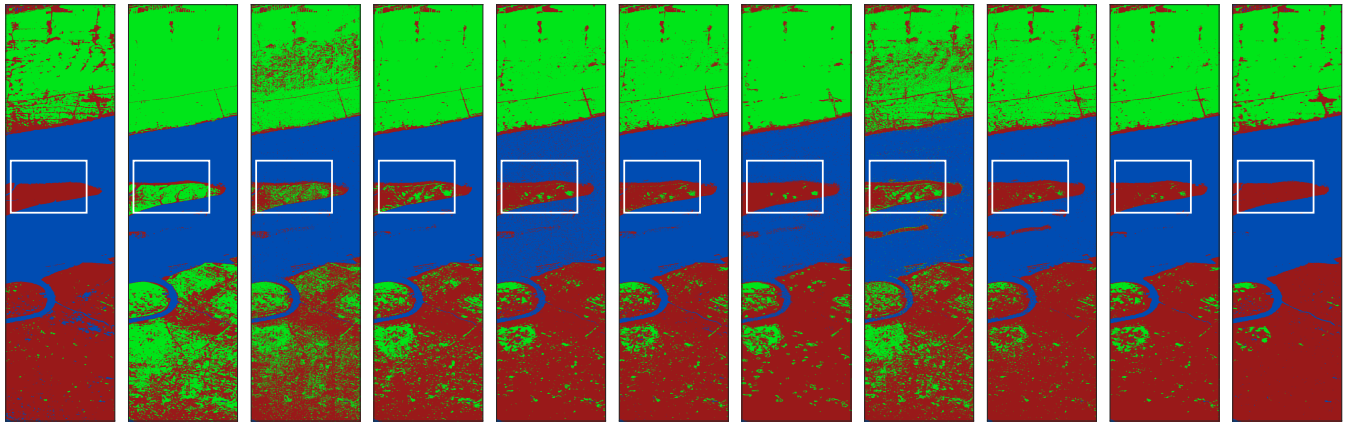
D. Ablation Analysis and Discussion

In this section, the effectiveness of TAADA is analyzed from the following aspects: ablation analysis, hyperparameter analysis, feature visualization, and computational time.

1) Ablation Analysis: To verify the effectiveness of each step of the proposed TAADA, we perform ablation studies on the Houston data, and the results are shown in Table VII. In this case, *M* stands for the baseline adversarial DA algorithm MCD, *D* represents our multilayer FC-BN-ReLU-Dropout structured discriminator, and *G* stands for the generator TBFE.

When only MCD is used, the OA is 74.49%. Although the network uses the idea of adversarial learning, it does not consider the characteristics of HSIs and the corresponding performance is not very well. When MCD is combined with TBFE, TBFE can be used as a generator to extract the attention-based spatial–spectral features of HSIs, and the classification accuracy of the algorithm is improved by 5.16%. When MCD is combined with the discriminator *D*, of which the classifier has a multilayer FC-BN-RELU-Dropout structure, the classifier can predict labels more accurately, and the accuracy rate is improved by 1.69%. When the three modules

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 8. Hangzhou–Shanghai task. (a) Ground truth, (b) KNN (72.07%), (c) SA (77.91%), (d) JDA (85.29%), (e) DAN (87.64%), (f) DSAN (89.99%), (g) MRAN (87.63%), (h) ADDA (81.00%), (i) MCD (86.32%), (j) TSTnet (90.64%), and (k) TAADA (94.17%). (The white rectangle circles indicate the areas that TAADA classifies more accurately compared with the comparison algorithm.)

TABLE VII
ABLATION STUDY ON THE HOUSTON TASK

Model	M	G	D	OA (%)
M	✓			74.49
M+G	✓	✓		79.65
M+D	✓		✓	76.18
TAADA	✓	✓	✓	**81.63**

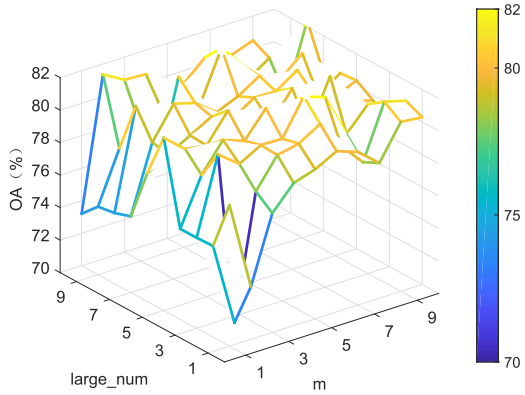


Fig. 9. OA versus parameters large_num and m on the Houston task.

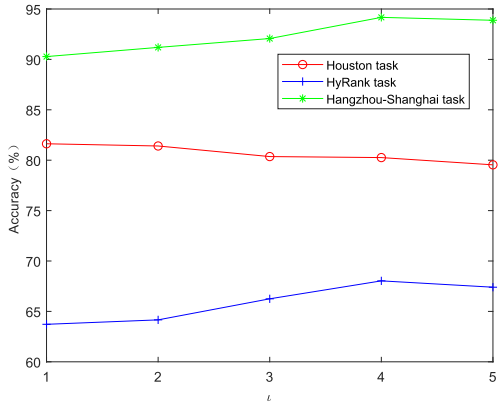


Fig. 10. OA versus the number of layers i on the three tasks.

TABLE VIII
ABLATION STUDY OF THE TWO-BRANCH FEATURE EXTRACTOR G ON THE HOUSTON TASK

Model	ResNet18	SE	Spectral	Spatial	OA (%)
ResNet18	✓				73.55
ResNet18+SE	✓	✓			73.91
Spectral			✓		79.26
Spectral+SE		✓	✓		79.47
Spatial				✓	77.60
Spatial+SE		✓		✓	77.78
Spatial+Spectral			✓	✓	79.32
TBFE		✓	✓	✓	**81.63**

are combined, they work the best. Finally, the prediction accuracy of TAADA in the Houston task reaches 81.63%. Experiments show that each module in the proposed TAADA contributes to the performance improvement.

As the TBFE-based feature generator contributes to the final performance improvement most, we further analyze the effect of different feature combinations in Table VIII, where the combination of baseline ResNet18, spectral, and spatial features with the SE-based attention module is considered. The discriminator of TAADA is unchanged and the generator is set as listed in Table VIII.

As can be seen from Table VIII, when the generator is a simple neural network ResNet18, the model is less effective due to the partial loss of the spatial and spectral information in the extracted features, and the accuracy of the model is only 73.91% even if the SE module is added again. When using the spatial and spectral branches alone, the model accuracy can reach 77.60% and 79.26%, and further adding the SE module to the branches slightly increases the accuracies. When combining the two branches but not using the SE module, an accuracy of 79.32% is achieved. Finally, TAADA achieves an accuracy of 81.63% by considering both the spatial–spectral features and SE modules. Experimentally, TBFE can extract the discriminative features despite its simple structure.

2) Parameter Analysis: In the proposed TAADA, the internal iteration parameter m and data augmentation parameter large_num need to be adjusted. m is a tradeoff parameter

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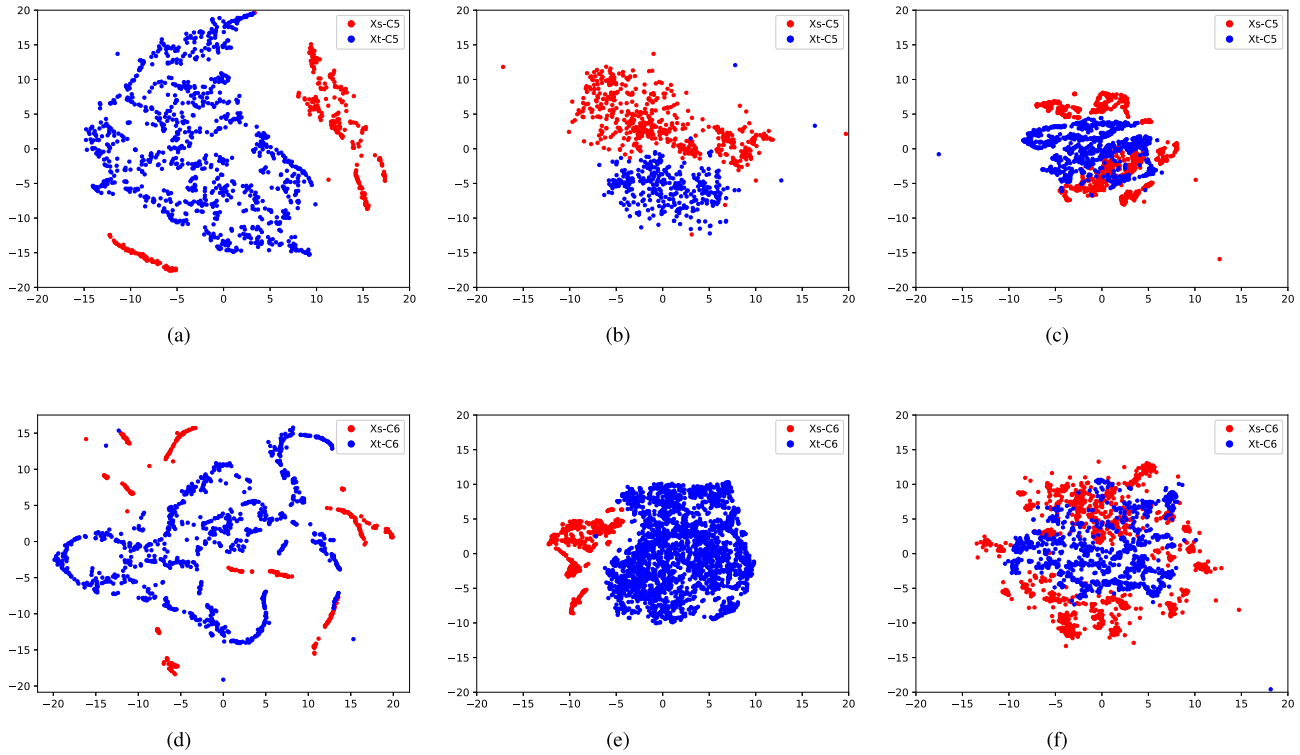


Fig. 11. tSNE-based alignment performance on the Houston task. (a) Original features of Houston (class 5). (b) ADDA features of Houston (class 5). (c) TAADA features of Houston (class 5). (d) Original features of Houston (class 6). (e) ADDA features of Houston (class 6). (f) TAADA features of Houston (class 6).

between the generator and the classifiers. large_num is used to preprocess the data. Before the data are put into the network, the data are augmented according to the value of large_num. This operation first shuffles the data, and then repeatedly splices the data according to the size of large_num, and finally the obtained data are large_num times the original data. Fig. 9 shows the OA of TAADA on the Houston task when the parameters m and large_num are varied simultaneously. It can be seen that the proposed TAADA method shows relatively better results in a large region of parameters. Taking into account the network training speed and classification performance, we set $m = 4$ and large_num $= 3$ in the experiments.

In TAADA, the classifier has a multilayer FC-BN-ReLU-Dropout structure. Fig. 10 shows the effect of the number of layers i on the final classification performance. With the increase in the number of layers, the OA on the HyRank and Hangzhou–Shanghai tasks is improved and reach the maximal at four layers. In the Houston task, different layers show similar results. Based on these results, the number of layers in the Houston, HyRank, and Hangzhou–Shanghai tasks is set as 1, 4, and 4, respectively.

In addition, we conduct an experimental analysis regarding the choice of window size for three datasets. The OA results of the proposed TAADA in different window sizes are shown in Table IX.

TABLE IX
OA (%) OF TAADA WITH DIFFERENT WINDOW SIZES

Window size	Houston	HyRank	Hangzhou-Shanghai
5×5	79.53	68.03	91.62
7×7	81.63	63.06	94.17
9×9	81.55	61.83	93.42

3) Feature Visualization: When using DA algorithms, it is critical for data from both the domains to be well-aligned. Figs. 11–13 show the aligned performance of TAADA on three DA tasks with the use of t-distributed stochastic neighbor embedding (tSNE). We illustrate the alignment performance using the fifth and sixth categories, because these two categories ("Residential buildings" and "Nonresidential buildings") are spectrally similar and hence difficult to be aligned. In addition, these two categories have large number of samples and may have significant within-class variations. Similarly, the seventh and eighth categories in the HyRank and the first and second categories in the Hangzhou–Shanghai task are selected to show the alignment performance. It can be seen from Figs. 11–13 that the original source and target data are obviously distributed in different regions [see (a) and (d) in each figure]. The traditional adversarial method ADDA aligns the source and target samples at a certain extent. However, as the domain classifier (i.e., discriminator) in ADDA only distinguishes the samples from which domains, it lacks the class discriminative ability. The source and target features aligned by ADDA still have different distributions, especially on the Houston task [see Fig. 11(b) and (e)]. After the TAADA feature alignment, the data of the source and target domains are well-aligned and approximately formed a single cluster together.

4) Computational Time: Here, we list the computational time of each algorithm on the Houston task. All the experiments are based on the PyTorch deep learning framework on

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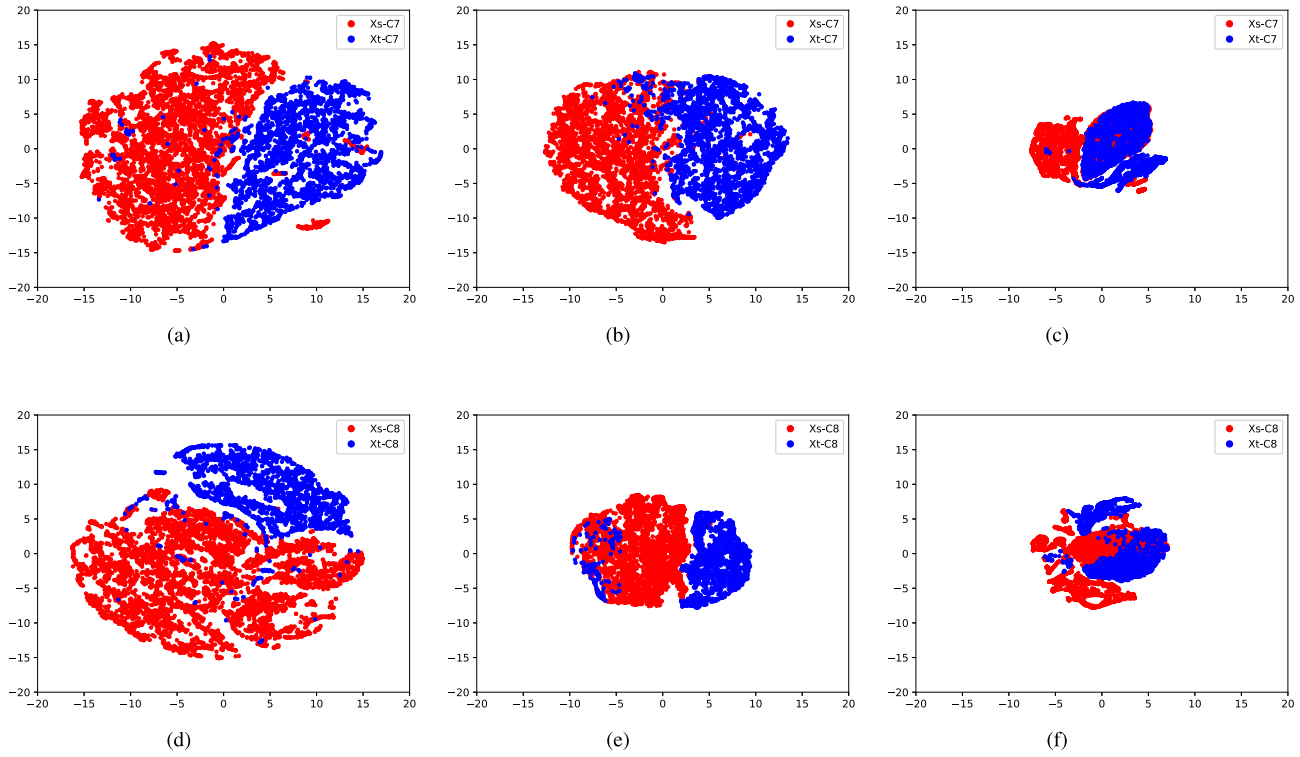


Fig. 12. tSNE-based alignment performance on the HyRank task. (a) Original features of HyRank (class 7). (b) ADDA features of HyRank (class 7). (c) TAADA features of HyRank (class 7). (d) Original features of HyRank (class 8). (e) ADDA features of HyRank (class 8). (f) TAADA features of HyRank (class 8).

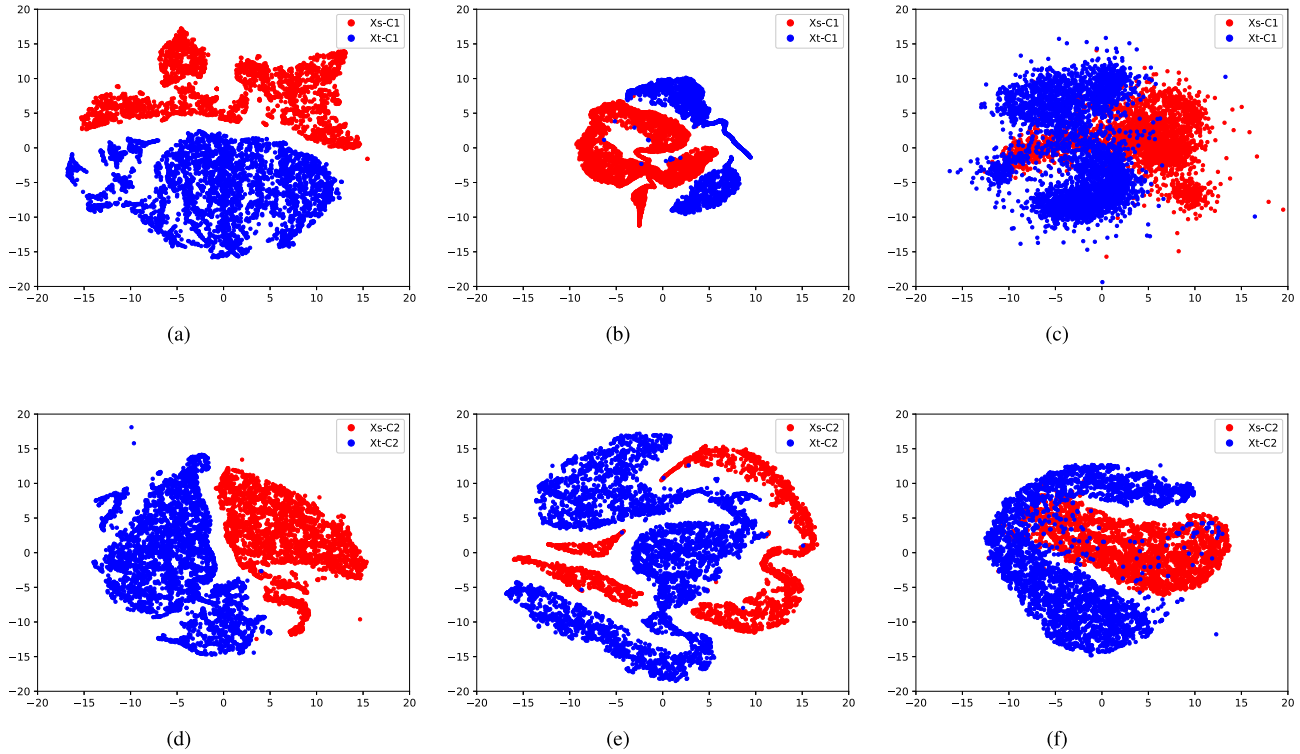


Fig. 13. tSNE-based alignment performance on the Hangzhou–Shanghai task. (a) Original features of Hangzhou–Shanghai (class 1). (b) ADDA features of Hangzhou–Shanghai (class 1). (c) TAADA features of Hangzhou–Shanghai (class 1). (d) Original features of Hangzhou–Shanghai (class 2). (e) ADDA features of Hangzhou–Shanghai (class 2). (f) TAADA features of Hangzhou–Shanghai (class 2).

a computer with a 2.7-GHz Intel Xeon Gold 6258R CPU and four NVIDIA GeForce RTX 2080 Ti GPUs. Table X shows the training time for one epoch with one forward propagation and one backward propagation, with a batch size of 128, and the training parameters and data settings are consistent with the previous experiments. The computational time of the proposed TAADA is slightly higher than the traditional adversarial methods and TSTnet.

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TABLE X
EXECUTION TIME (IN SECONDS) FOR ONE EPOCH TRAINING OF DIFFERENT METHODS ON THE HOUSTON TASK

Data	DAN	DSAN	MRAN	ADDA	MCD	TSTnet	TAADA
Houston	4.63	5.28	4.89	4.69	3.24	6.58	8.65

IV. CONCLUSION

In this article, we have proposed a TAADA network for cross-domain HSI classification. TAADA first uses TBFE to extract the attention-based spatial–spectral joint features of HSIs, and then uses the idea of adversarial training between two classifiers and a generator to align the distribution of the source and target domains. Finally, the multilayer FC-BN-ReLU-Dropout structure classifier is used to improve the classification performance. The results on three experiments show that TAADA can effectively perform cross-domain classification of HSIs and outperforms other DA techniques.

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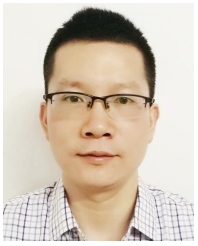
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